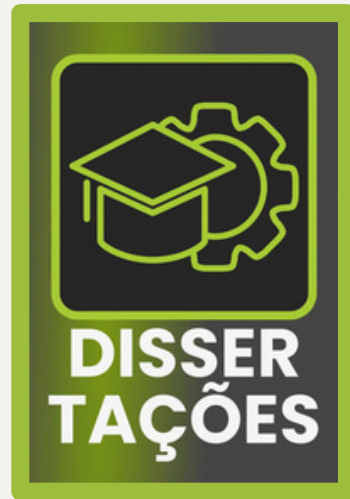




DSD – M3: Legal requirements, technical and business risks.

Projeto em Engenharia
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Presented by
Group 16

Our team

João Ferreira

jd.ferreira@ua.pt

Giovanni Santos

giovanni.s@ua.pt

Christian Fernandes

christian.fernandes@ua.pt



Guilherme Mesquita

g.mesquita07@ua.pt

Regina Tavares

regina.tavares@ua.pt

Rafael Semedo

rafaeljcmsemedo@ua.pt

Advisor:

Diogo Gomes

dgomes@ua.pt

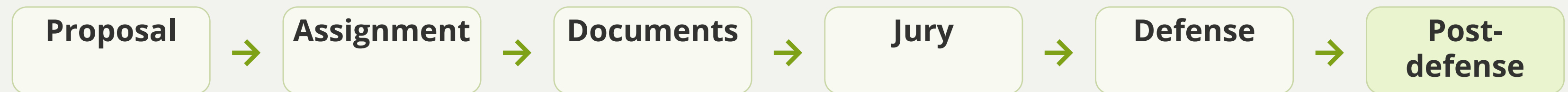
Project context

- DSD continues the existing platform at dissertacoes.av.it.pt.
- It supports the dissertation / project / internship lifecycle.
- Current work extends defense and post-defense document workflows.

M3 analyses the whole DSD platform, using defense document management as the newest and most legally/technically sensitive example.

Platform scope

DSD supports the academic workflow from proposal to post-defense documentation.



- Student interests, course validation and assignment connect early-stage decisions.
- Document submission, jury homologation and defense scheduling are operational checkpoints.
- Metrics and quotas provide administrative visibility across the workflow.

Actors and responsibilities

Each profile has different responsibilities and access permissions.

Profile	Main responsibility in DSD
Student	Interests, assignment, document submission
Teacher / Advisor	Proposals, supervision, jury proposal
Course Director	Course validation, jury nomination, quotas
Staff / Secretary	Document validation, scheduling, homologation
Admin	Platform configuration and governance
Jury roles	Post-defense documents and signatures

Legal requirements

GDPR & Data Protection

- Personal academic data
- Defined purpose & retention
- Data minimization

Academic context

- Institutional validation
- Academic regulations

Security & Compliance

- Role-based access (RBAC)
- Audit logs
- Strong digital signatures (eIDAS Framework)

Licensing & Terms

- Apache 2.0 (project)
- Permissive dependencies (MIT, BSD)
- No GPL risks

Technical risks and mitigations

Risk	Impact on DSD	Mitigation
Dispersed RBAC	Wrong access to sensitive actions/documents	Backend permission checks and endpoint review
Document storage	Retention, backup and access risk	Dedicated storage, metadata in Mongo, backups
External dependencies	Login, email or institutional services unavailable	Fallback procedures and clearer error states
Digital/manual signatures	Invalid or incomplete official documents	Validation rules + manual acceptance path + audit trail
Low automated tests	Workflow regressions in critical paths	Tests for critical endpoints and workflows

Data protection strategy

The legal requirements map directly to technical controls in the DSD architecture.

Metadata minimization

Store only necessary metadata in Mongo.

Document storage

Keep files in dedicated storage with access control.

Retention

Define rules for old dissertations, documents and logs.

Backend RBAC

Protect sensitive actions at endpoint level.

Auditability

Keep audit trails for document decisions.

Signatures

Store only the minimum signature validation data needed for audit.

SWOT analysis

Strengths

- Complete academic workflow
 - Real institutional roles
- Less manual email/document handling
 - Quotas and operational visibility

Custom-made for DETI – suited to their needs

Opportunities

- Extend to more courses/departments
 - Better traceability of decisions
 - Reduce paper-based processes
- Future exports to official systems

Not everyone at DETI believes Fenix will suit their needs any time soon

Weaknesses

- Legacy/top-level document fields
- RBAC repeated across endpoints
 - Limited test coverage
- Dependence on UA authentication, email and storage services

Not easily applicable to other universities

Threats

- Retention policy not fully defined
 - Authentication/email outages
 - Academic rules may change
- Signature support may be incomplete

Strong competition from bigger, reputable teams (FenixEdu)

TOWS strategies

Concrete strategies derived from the DSD SWOT.

SO

Use full workflow coverage to promote DSD as an institutional academic management tool.

WO

Address document-model debt through a phased refactoring and testing roadmap.

ST

Strengthen audit logs and RBAC to reduce legal and operational risks.

WT

Define retention, backups and manual fallback procedures.

PESTEL

External context for DSD as an academic workflow platform.

Political

Aligns with UA administrative digitalization

Economic

Reduces manual coordination time

Social

Improves transparency, but requires adoption effort

Technological

Depends on auth, email, databases and document validation

Environmental

Reduces paper in academic workflows

Legal

GDPR, retention, signatures and academic regulations

Next steps

Planned actions to reduce risk before broader adoption.

- Finalize signature flow, including manual signatures.
- Define document and log retention policy.
- Consolidate permission rules by profile.
- Add tests for critical workflows.
- Review dependency licenses / SBOM.
- Prepare MS4 validation plan for critical workflows, security and scalability

Thank you!



 <https://pei-dsd-2025.github.io/microsite/> 